

Randomized and low-rank approximation

Random column subsets of arbitrary fixed-sparsity matrices

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because arbitrary support intersections defeat existing sparse-matrix estimates; community impact is understanding limits of sparse sketches and column selection.

Problem statement

Fix an integer $s \geq 2$ and $C \geq 1$. Suppose $r_k \leq k$, $k/r_k \rightarrow C$, and $n_k/r_k \rightarrow \infty$. For every deterministic sequence $M_k \in \{-1, 0, 1\}^{k \times n_k}$ with exactly s nonzero entries per column, and a uniformly random r_k -element column set I_k , prove or disprove

$$\forall \eta > 0, \quad \Pr\{\sigma_{\min}((M_k)_{:I_k}) > \eta\} \rightarrow 0.$$

Here $\sigma_{\min}(B) = \inf_{\|x\|_2=1} \|Bx\|_2$. There is no hypothesis controlling intersections of column supports.

Reference

Han Huang, Mark Rudelson, and Konstantin Tikhomirov, *Well-Invertible Column Subsets of Sparse Matrices Are Rare*, §7, Conjecture 7.1; compare Theorem 1.3 for the additional structural hypothesis in the proved result.

Status check

Searches included “*unrestricted deterministic sparse sketches*” conjecture and the exact paper title with 2026. No subsequent resolution was located. The paper’s disproof of a particular SparseStack conjecture does not establish its own unrestricted deterministic conjecture.

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Rechecked [Huang–Rudelson–Tikhomirov, Theorem 1.3 and Conjecture 7.1](#): a structural subclass is proved, while arbitrary deterministic supports remain conjectural. Title and sparse-subset follow-up searches found no full resolution. [Tikhomirov’s later embedding paper](#) studies random models and spectral-norm estimates, not all deterministic support patterns.